

① Piezo1 $\xrightarrow{\text{in}}$ Keratinocytes $\xrightarrow{\text{in}}$ skin.

show expression.

(CKO) mouse. $\sim$ wildtype (WT) $\sim$

$\uparrow$ compare wound closure rate.
faster

(GOF)
gain-of-function
mutation.

$\uparrow$
slower

conclusion: Piezo knockout accelerates wound closure.

②

cellular traction forces $\xrightarrow{\text{generate}}$ Piezo1

$\downarrow$ very important for
Keratinocytes migration

?
may play a role

test: ~~display~~ display Piezo1 Ca^{2+} flickers

③ experiment allow us to see the sub-cellular ~~location~~ localization.

↑
is dynamic :

in individual Keratinocytes its enriched.

in the retracting, rear end of migrating cell

check the monolayer of Keratinocytes



↑
with scratched.

to model the wound healing process in vitro

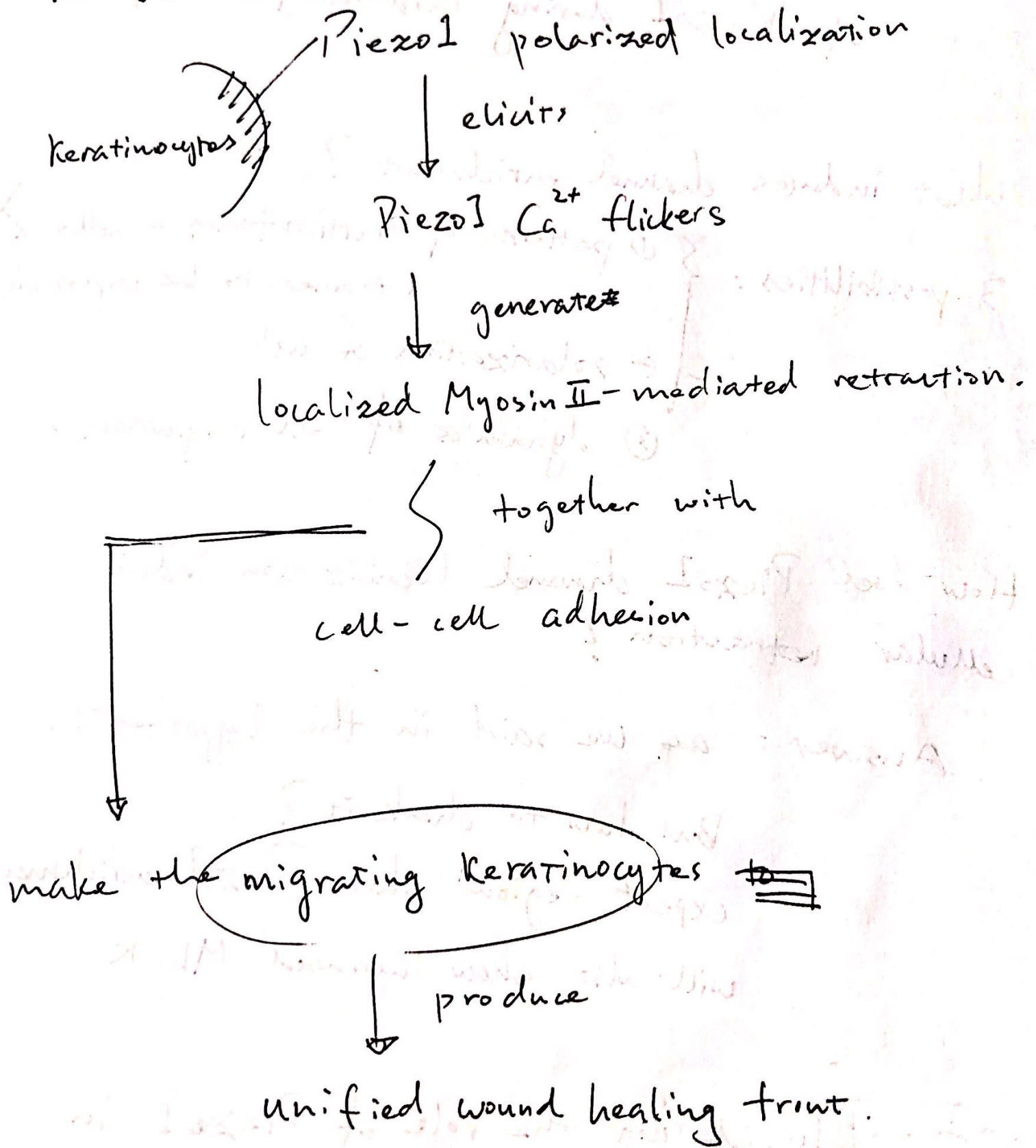
experiment: using time-lapse imaging

Piezo1, ~~is~~ precedes the.

← retraction of the region.

conclusion: Piezo1 plays a role in cellular retraction event.

Hypothesis



Aim I. determine the spatiotemporal dynamics of Piezo1 during keratinocytes migration

• What induces channel enrichment?

3 possibilities: { ① patterns of traction forces in cells (examined to be impossible) X
② polarization of cells
③ dynamics of cell migration.

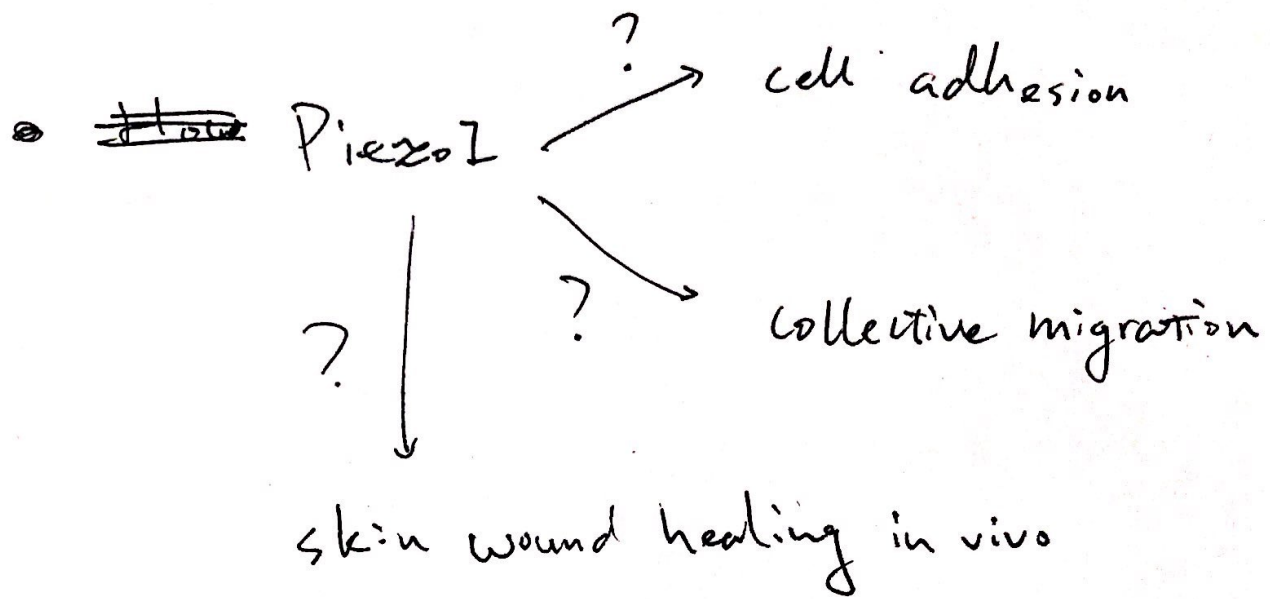
• How does Piezo1 channel localization induce cellular retraction?

Answer: as we said in the hypothesis.

But how to check it?

expert regions show Piezo1 enrichment
will also show increased MLCK

Aim II. Elucidating the role of Piezo1 in keratinocyte migration and skin wound healing



Aim III. Math Modeling